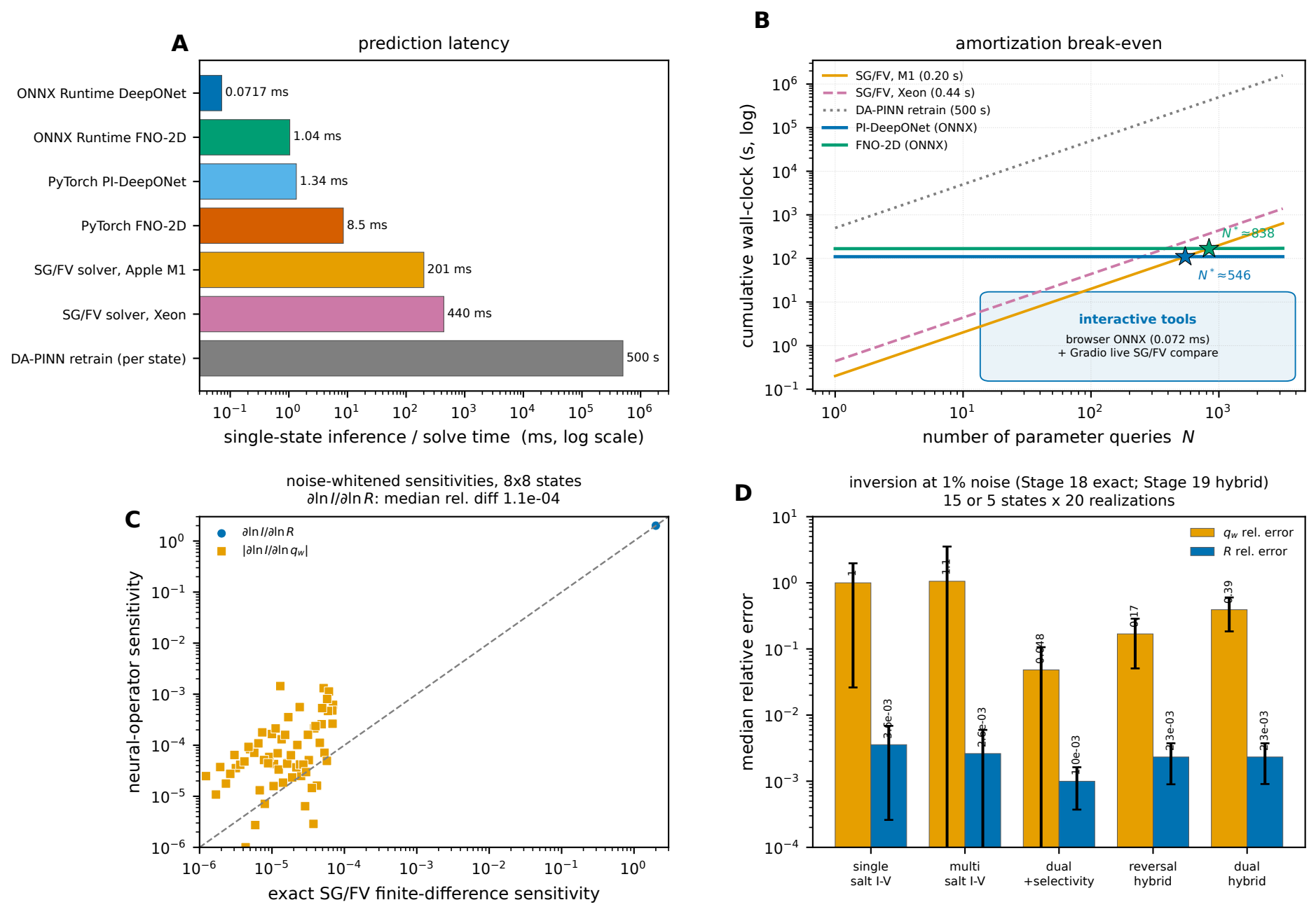
A
prediction latency
ONNX Runtime DeepONet
ONNX Runtime FNO-2D
PyTorch PI-DeepONet
PyTorch FNO-2D
SG/FV solver, Apple M1
SG/FV solver, Xeon
DA-PINN retrain (per state)
0.0717 ms
1.04 ms
1.34 ms
8.5 ms
201 ms
440 ms
500 s
single-state inference / solve time (ms, log scale)
B
amortization break-even
SG/FV, M1 (0.20 s)
SG/FV, Xeon (0.44 s)
DA-PINN retrain (500 s)
PI-DeepONet (ONNX)
FNO-2D (ONNX)
cumulative wall-clock (s, log)
N* ≈ 838
N* ≈ 546
interactive tools
browser ONNX (0.072 ms)
+ Gradio live SG/FV compare
number of parameter queries N
C
noise-whitened sensitivities, 8x8 states
∂ln I/∂ln R: median rel. diff 1.1e-04
∂ln I/∂ln R
|∂ln I/∂ln q_w|
neural-operator sensitivity
exact SG/FV finite-difference sensitivity
D
inversion at 1% noise (Stage 18 exact; Stage 19 hybrid)
15 or 5 states x 20 realizations
q_w rel. error
R rel. error
median relative error
1
3.6e-03
1.1
2.6e-03
0.048
1.0e-03
0.17
3.3e-03
0.39
3.3e-03
single salt I-V
multi salt I-V
dual +selectivity
reversal hybrid
dual hybrid